DC Motor Speed Control using PWM and Arduino

Ayush Bhardwaj (23BEE0118)
Department of Electrical and Electronics Engineering
Vellore Institute of Technology, Vellore, India

Karan Pawar (23BEE0038)
Department of Electrical and Electronics Engineering
Vellore Institute of Technology, Vellore, India

Furqan Ansari (23BEE0081)
Department of Electrical and Electronics Engineering
Vellore Institute of Technology, Vellore, India

***Abstract*—This paper presents the design and implementation of a DC motor speed and direction control system using an Arduino microcontroller, L293D motor driver, and a potentiometer. The proposed system allows the user to vary motor speed and direction efficiently. The potentiometer provides analog input to the Arduino, which generates corresponding PWM signals to control the motor driver. The system demonstrates efficient, cost-effective, and user-friendly motor control suitable for small-scale automation projects and educational applications.**

***Index Terms*—Arduino, DC Motor, L293D, PWM, Potentiometer, Speed Control.**

I. Introduction

DC motors are widely used in applications such as robotics, electric vehicles, and industrial drives because of their precise speed control capabilities. Controlling motor speed and direction efficiently is essential in automation systems. Conventional controllers are complex, but microcontroller-based designs offer flexibility and simplicity. This project demonstrates speed and direction control of a DC motor using an Arduino UNO, an L293D motor driver, and a potentiometer that generates a variable PWM signal.

II. Literature Review

Liu et al. (2019) studied various PWM techniques for DC motor control. Jain et al. (2020) implemented an Arduino-based speed control system for educational purposes. Kumar et al. (2018) designed a low-cost analog-based control system. Zhang et al. (2017) explored PID-based motor control for better precision. This project emphasizes a simple, low-cost method with real-time manual control through a potentiometer.

III. Hardware Components

- **Arduino UNO:** Reads analog voltage and generates PWM signals.
- **L293D Motor Driver:** Provides bidirectional control to the motor.
- **Potentiometer:** Acts as a user input to vary the voltage from 0–5V.
- **DC Motor:** The load whose speed is being controlled.
- **Power Supply:** Provides power to Arduino and motor driver.

S. No.	Component Name	Specification / Type	Quantity	Purpose / Description
1	**Arduino UNO**	ATmega328P based microcontroller board	1	Main controller — reads potentiometer, controls motor and LEDs
2	**L293D Motor Driver IC**	Dual H-Bridge driver	1	Drives DC motor forward/reverse via Arduino signals
3	**DC Motor**	6–12 V brushed DC motor	1	Load for speed/direction control
4	**Potentiometer**	10 kΩ linear	1	Used to control speed and direction (analog input to A0)
5	**LED – Green**	5 mm	1	Indicates **Forward** direction
6	**LED – Red**	5 mm	1	Indicates **Reverse** direction
7	**Resistors**	220 Ω or 330 Ω (not kΩ)	2	Current-limiting resistors for LEDs
8	**Breadboard**	Standard size	1	For circuit prototyping
9	**Jumper Wires (M–M)**	Assorted colors	~15	For connecting components on breadboard
10	**Power Supply for Motor**	6 V or 9 V battery / adapter	1	Provides motor voltage to L293D pin 8 (Vcc2)
11	**USB Cable**	Type A to B	1	For uploading code & powering Arduino
12	**Common Ground**	—	—	Connect Arduino GND, L293D GND, and power supply negative together

Fig. 1. List of hardware components with specifications and purposes.

IV. Circuit Diagram

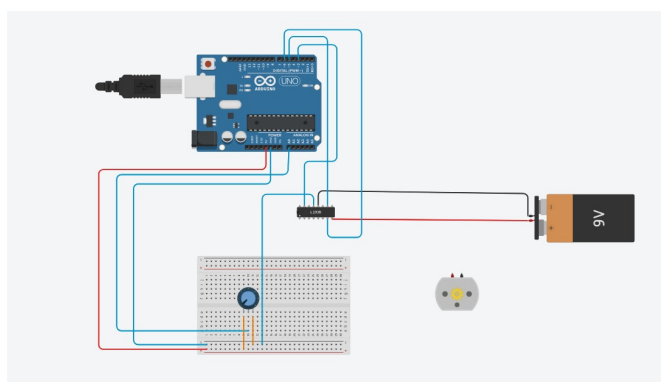


Fig. 2. Circuit diagram of the DC motor speed control system.

V. Methodology

The overall design flow is shown in Fig. 3. The potentiometer provides an analog voltage input to Arduino, which reads it via an ADC pin. The Arduino maps the voltage to a PWM signal, which is used to drive the motor through the L293D driver IC. The rotation direction changes based on the potentiometer's position with respect to the midpoint.

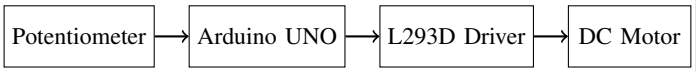


Fig. 3. Block Diagram of the DC Motor Speed Control System.

VI. Algorithm

1) Initialize motor control and potentiometer pins.
2) Read analog input from potentiometer (0–1023).
3) Map input to PWM range (0–255).
4) If input < 512, rotate motor reverse with proportional speed.
5) If input > 512, rotate motor forward with proportional speed.
6) If input ≈ 512, stop motor.
7) Repeat continuously.

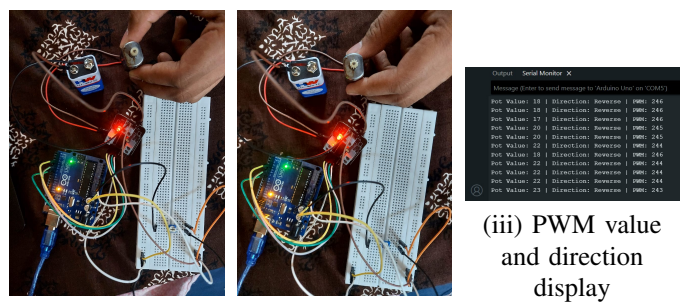
VIII. Arduino Code

Listing 1. Arduino code for DC motor speed and direction control using potentiometer

```
const int potPin = A0;
const int in3 = 5;
const int in4 = 3;
const int en2 = 6;

int potValue = 0;
int motorSpeed = 0;
String direction = "Stop";

void setup() {
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);
  pinMode(en2, OUTPUT);
  Serial.begin(9600);
  Serial.println("DC Motor Control Initialized...");
}

void loop() {
  potValue = analogRead(potPin);
  int deadZoneLow = 480;
  int deadZoneHigh = 540;

  if (potValue > deadZoneHigh) {
    // Forward direction
    digitalWrite(in3, HIGH);
    digitalWrite(in4, LOW);
    motorSpeed = map(potValue, deadZoneHigh, 1023,
        0, 255);
    analogWrite(en2, motorSpeed);
    direction = "Forward";
  }
  else if (potValue < deadZoneLow) {
    // Reverse direction
    digitalWrite(in3, LOW);
    digitalWrite(in4, HIGH);
    motorSpeed = map(potValue, 0, deadZoneLow, 255,
        0);
    analogWrite(en2, motorSpeed);
    direction = "Reverse";
  }
  else {
    // Stop
    digitalWrite(in3, LOW);
    digitalWrite(in4, LOW);
    analogWrite(en2, 0);
    direction = "Stop";
    motorSpeed = 0;
  }

  Serial.print("Pot Value: ");
  Serial.print(potValue);
  Serial.print(" | Direction: ");
  Serial.print(direction);
  Serial.print(" | PWM: ");
  Serial.println(motorSpeed);

  delay(200);
}
```

VII. Results and Discussion

The motor speed changed linearly with potentiometer rotation. When the potentiometer knob was in the mid-position, the motor stopped. Turning clockwise increased speed in the forward direction, while counter-clockwise rotation caused reverse motion. PWM output from Arduino ensured smooth speed variation without jerks. The overall setup consumed less power and operated reliably.



(i) Motor in reverse direction (ii) Motor not in motion (iii) PWM value and direction display

Fig. 4. Experimental setup showing (i) motor in reverse direction, (ii) stationary motor, and (iii) PWM value with direction indication.

VIII. Conclusion

A simple and cost-effective DC motor speed and direction control system was implemented using Arduino and the L293D motor driver. The system achieved efficient control and can be enhanced by incorporating PID feedback, sensors, or Bluetooth-based remote control.

IX. References

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